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Tangeretin offers neuroprotection against colchicine-induced memory impairment in Wistar rats by modulating the antioxidant milieu, inflammatory mediators and oxidative stress in the brain tissue

Olalekan Bukunmi Ogunro^{1*}, Mojisola Esther Karigidi¹, Gideon Ampoma Gyebi², Areej Turkistani³ and Ahmad H. Almehmadi⁴

Abstract

Background Tangeretin, a flavone compound (O-polymethoxylated) naturally present in tangerine and other citrus peels has demonstrated effectiveness as an anti-inflammatory and neuroprotective agent in several disease model. This study evaluated the impact of tangeretin in mitigating cognitive dysfunction and oxidative stress induced by colchicine in rats, comparing its efficacy with donepezil hydrochloride.

Methods Cognitive dysfunction was induced by administering colchicine (15 µg/rat) intracerebroventricularly (ICV) via a stereotaxic apparatus in male Wistar rats. Colchicine resulted in poor memory retention in acquiring and retaining a spatial navigation task, passive avoidance apparatus, and Morris water maze paradigms. Chronic treatment with tangeretin (at doses of 50, 100, and 200 mg/kg, p.o. once daily) and donepezil hydrochloride (at a dose of 10 mg/kg, p.o. daily) for 28 days, starting seven days before colchicine injection, significantly ameliorated colchicine-induced cognitive impairment.

Results The biochemical analysis showed that chronic administration of tangeretin effectively reversed the colchicine-induced increase in the level/activity of lipid peroxidation, hydrogen peroxide (H₂O₂), myeloperoxidase (MPO), nitrite, reactive oxygen species (ROS), tumour necrosis factor-α (TNF-α), nuclear factor kappa B (NF-κB), interleukin-1β (IL-1β), interleukin-6 (IL-6), interleukin-10 (IL-10), serotonin, dopamine, glutamate, amyloid beta (Aβ) peptide, and caspase-3. Tangeretin also reversed the colchicine-induced reduction in the level/activity of brain-derived neurotrophic factor (BDNF), gamma-aminobutyric acid (GABA), acetylcholinesterase (AChE), glutathione S-Transferase (GST), glutathione peroxidase (GPx), glutathione reductase (GR), catalase (CAT), superoxide dismutase (SOD), reduced glutathione (GSH), and total thiol (T-SH) in rat brains. However, donepezil hydrochloride did not prevent oxidative stress.

*Correspondence:

Olalekan Bukunmi Ogunro
olalekan.ogunro@koladaiiuniversity.edu.ng;
olalekanbukunmi@gmail.com

Full list of author information is available at the end of the article



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Conclusions These findings suggest that chronic administration of tangeretin at 50, 100, and 200 mg/kg, p.o. once daily, was protective in mitigating colchicine-induced cognitive impairment and associated oxidative stress. At the same time, donepezil hydrochloride did not demonstrate similar effects.

Keywords Alzheimer's disease, Neurodegenerative diseases, Dementias, Cognition disorder, Neurotransmitter, Oxidative stress

Background

Alzheimer's disease (AD) is a neurological illness that worsens with time, impacting behaviour, memory, and cognitive abilities [1]. It is the most common cause of dementia in the elderly, affecting millions worldwide. As the population ages, the prevalence of AD is expected to rise, posing challenges to healthcare systems and society [2, 3]. While there is no cure for AD, several FDA-approved medications like cholinesterase inhibitors (e.g., donepezil) and memantine can only manage symptoms in early to moderate stages [4]. Additionally, new treatments like lecanemab and donanemab are showing promise in addressing the underlying biology of the disease, but they still have their side effects [5, 6]. Non-drug interventions such as cognitive training, diet, physical activity, and creating a supportive environment, which is also crucial in managing AD, may also not address the main problem [7]. Alternative treatments like vitamin E, curcumin, and omega-3 fatty acids have been studied but lack strong evidence for treating AD symptoms. Insights into the disease and developing effective treatments are essential for addressing the public health impact of Alzheimer's disease, especially as it relates to memory impairment and cognitive decline.

Memory impairment and cognitive decline are standard features of various neurodegenerative conditions, including Alzheimer's disease and related dementias. These conditions pose significant challenges to both affected individuals and healthcare systems worldwide. Consequently, there is a growing interest in exploring potential interventions that could mitigate memory deficits and protect against neurodegenerative damage [8, 9].

Colchicine, a naturally occurring compound derived from the autumn crocus (*Colchicum autumnale*), has been employed in test subjects to cause memory impairment because of its microtubule-disrupting properties [10, 11]. Among the effects it has on the central nervous system are interference with axonal transport, neural inflammation, and oxidative stress, all of which contribute to cognitive dysfunction [12].

Tangeretin, a flavonoid compound found primarily in citrus fruits, has garnered attention recently for its potential neuroprotective properties. Previous studies have suggested that tangeretin possesses antioxidant, anti-inflammatory, and anti-apoptotic characteristics [13] that make it a promising candidate for alleviating memory deficits and preventing neural damage [14]. The

present research aims to investigate whether tangeretin can offer neuroprotection against the colchicine-induced rat model of memory impairment.

Numerous in vitro or in vivo studies have demonstrated that tangeretin exhibits potent antioxidant activity, capable of scavenging free radicals and enhancing the activity of endogenous antioxidant enzymes [15, 16]. For instance, Kou et al. [17, 18] reported that tangeretin increased the levels of superoxide dismutase (SOD) and glutathione peroxidase (GPx) in neuronal cell lines subjected to oxidative stress, thereby reducing cellular damage. This ability to modulate the antioxidant defence system is critical, as oxidative stress is a crucial contributor to neuronal cell death and cognitive decline in various neurodegenerative diseases, including Alzheimer's disease (AD) and related dementias. In addition to its antioxidant effects, tangeretin has been shown to exhibit significant anti-inflammatory activity. In a study by Shu et al. [19], tangeretin reduced the production of pro-inflammatory cytokines such as TNF- α and IL-6 in microglial cells activated by lipopolysaccharide (LPS). This reduction in inflammatory mediators is particularly relevant in neurodegeneration, where chronic inflammation contributes to neuronal damage and cognitive impairments. By modulating the inflammatory response, tangeretin may help mitigate the progression of neurodegenerative diseases [20].

Tangeretin's neuroprotective potential has also been explored in animal models of neurodegeneration [21]. For instance, a study by Hung et al. [22] demonstrated that tangeretin significantly improved cognitive function in a cultured primary rat model of Alzheimer's disease induced by amyloid-beta (A β) oligomers. The treatment was associated with reduced oxidative stress markers and decreased levels of inflammatory cytokines in the brain. This supports the hypothesis that tangeretin can mitigate the neurodegenerative processes through its combined antioxidant and anti-inflammatory effects. Another study by Datla et al. [23] investigated the effects of tangeretin in a rat model of Parkinson's disease induced by 6-hydroxydopamine (6-OHDA). The results showed that tangeretin treatment not only improved motor function but also preserved dopaminergic neurons in the substantia nigra. This neuroprotection was attributed to tangeretin's ability to reduce oxidative stress and inhibit neuroinflammation, further reinforcing its potential as a therapeutic agent in neurodegenerative conditions [24].

Given tangeretin's demonstrated ability to modulate oxidative stress and inflammatory mediators, it is hypothesised that this compound may offer neuroprotection against colchicine-induced cognitive deficits. The evidence strongly suggests that tangeretin could counteract the pathological processes triggered by colchicine in the brain. Tangeretin may protect neuronal integrity and improve cognitive function in this model by enhancing the antioxidant defence system and reducing neuroinflammation. This hypothesis is supported by previous *in vitro* and *in vivo* findings, which highlight tangeretin's potential to mitigate neurodegenerative processes through its dual antioxidant and anti-inflammatory actions.

Scientific insight into the impact of tangeretin on cognitive function and its mechanisms of neuroprotection in the context of colchicine-induced memory impairment is crucial for elucidating potential therapeutic interventions for memory-related disorders. The study provides information on comprehensive biochemical parameters like oxidative stress markers [Reactive oxygen species (ROS) concentrations, antioxidant enzymes including catalase (CAT), glutathione peroxidase (GPx), and superoxide dismutase (SOD), and lipid peroxidation products such malondialdehyde (MDA)]; neuroinflammatory markers [pro-inflammatory cytokines, including interleukin-1 β (IL-1 β) and tumour necrosis factor- α (TNF- α), anti-inflammatory cytokines, such as interleukin-10 (IL-10),]; neurotrophic factors [neurotrophic factors, such as nerve growth factor (NGF) and brain-derived neurotrophic factor (BDNF)]; neurotransmitters (levels of critical neurotransmitters involved in memory and cognitive function, such as serotonin, dopamine GABA and glutamate; apoptosis markers (Caspase-3 activity); neurotrophic factors [Brain-Derived Neurotrophic Factor (BDNF), Amyloid Beta (Ab) Peptide]; inflammatory mediators [myeloperoxidase (MPO) activity, nitric oxide (NO) levels. These biochemical parameters can provide insights into how tangeretin may exert its neuroprotective effects and mitigate colchicine-induced memory impairment in Wistar rats. By measuring these parameters, researchers can assess the impact of tangeretin on oxidative stress, neuroinflammation, apoptosis, neurotransmitter balance, and overall neuronal health.

This study, therefore, aims to contribute valuable insights to the field of neuroprotection and cognitive health, opening doors to novel strategies for managing and potentially preventing cognitive decline associated with neurodegenerative conditions.

Materials and methods

Chemicals

Acetylthiocholine iodide, donepezil hydrochloride, colchicine, glutathione, 5,5'-dithiobis (2-nitrobenzoic acid)

(DTNB), tangeretin (TGN, CAS No: 481–53–8/purity, 99%), epinephrine, hydrogen peroxide, 1-chloro-2,4-dinitrobenzene (CDNB), 2-thiobabutaric acid (TBA), bovine serum albumin (BSA), and thiobarbituric acid (TCA) was obtained from Sigma–Aldrich, USA.

Enzyme-Linked Immunosorbent Assay (ELISA) kits for TNF- α , NF- κ B, IL-1 β , IL-6, and IL-10; caspase-3; Brain-Derived Neurotrophic Factor (BDNF); amyloid beta (Ab) peptide were respectively procured from the Institute of Biological Engineering of Nanjing Jiancheng, Nanjing, China; Elabscience Biotechnology Company, Beijing, China; Elabscience Biotechnology Company, Beijing, China; and Boster Biological Technology, Fremont, California, USA; WKEA Med Supplies Corp., New York, USA (rat b-amyloid 1–42). Other reagents and chemicals were of the highest grade available.

Animals

Male Wistar rats, sourced from the Animal House of the Department of Biological Sciences at KolaDaisi University in Ibadan, weighing between 200 and 220 g at the onset of the surgical procedures, were utilised for the study. The animals underwent an acclimatisation period to the laboratory environment before the experiments. Post-surgery, these animals were housed in pairs in plastic cages with comfortable bedding, maintaining standard light-dark cycles, and provided *ad libitum* access to food and water.

All experiments were conducted between 08:00 and 16:00. The research procedure was approved by Ajayi Crowther University's research ethics committee (FNS/ERC/2022/020B) and followed the National Academy of Science's (NAS) recommendations for the moral use and appropriate handling of laboratory animals.

Surgery and intracerebroventricular (ICV) administration of colchicine

With a few minor adjustments, the surgical process adhered to a predetermined protocol. All animals were placed in a stereotaxic device and given an intraperitoneal dose of sodium pentobarbital (45 mg/kg). To insert the injection cannula into the lateral cerebral ventricle, a midline sagittal incision was created in the scalp, and two holes were drilled into the skull. The scalp was sutured after that. The method was the same in rats receiving the sham operation, except for the cannula implantation and hole drilling. The injection site was located 3.6 mm below the cortical surface, 1.8 mm lateral to the sagittal suture, and 0.8 mm posterior to the bregma [10].

To avoid infection, gentamicin (5 mg/kg, intraperitoneal) was administered to every animal after surgery. Gentamicin was selected for its well-documented ability to induce oxidative stress and neuroinflammation, providing a reliable model to assess the neuroprotective

effects of tangeretin. Intracerebroventricular (ICV) infusions of rats were made with either physiological saline or a solution that included 15 µg of dissolved colchicine in physiological saline. A Hamilton micro syringe placed within the injection cannula was used to inject these solutions (5 µl). The microsyringe was left in place for two minutes following the injection to promote diffusion. Throughout the recovery phase, extra care was taken to ensure the rat's cage had food and water.

Drugs and treatment protocol

Fresh solutions of tangeretin and colchicine were prepared for each experiment. The colchicine solution in physiological saline was prepared to deliver a dose of 15 µg in a 5 µl volume for intracerebroventricular administration. For oral intake, tangeretin and donepezil hydrochloride were suspended in physiological saline and given 1 ml per 100 g of body weight. The animals were randomly allocated into six groups ($n = 10$).

Group A: As the control group, this group of healthy rats underwent an intracerebroventricular injection (ICV) of physiological saline. Group B: Rats given a bilateral ICV injection of 15 µg/rat of colchicine and physiological saline to develop Alzheimer's disease (AD). Group C: The rats were given an oral dose of donepezil hydrochloride (10 mg/kg) 7 days before receiving a colchicine injection (15 µg/rat, ICV), and then they were given another 21 days of donepezil hydrochloride post-treatment. Group D: The rats were given an injection of colchicine (15 µg/rat, ICV) after 7 days of pre-treatment with tangeretin (50 mg/kg, once daily orally), and then

they were given another 21 days of post-treatment with the same dose of tangeretin. Group E: Rats were given a pre-treatment of tangeretin (100 mg/kg, once daily orally) for 21 days, then an ICV injection of colchicine (15 µg/rat), and finally a post-treatment of the same dose of tangeretin for another 21 days. Group F: The rats were given an injection of colchicine (15 µg/rat, ICV) after 7 days of pre-treatment with tangeretin (200 mg/kg, once daily orally), and then they were given another 21 days of post-treatment with the same dose of tangeretin (Fig. 1).

Based on previously published research, the dosages of were chosen for tangeretin [14], colchicine [25], and donepezil hydrochloride [26]. The study's duration was founded on the prior reports of Mohamed et al. [27] and Khurana et al. [25].

Behavioural assessment

Assessment of cognitive performance

The Morris water maze was used to assess the rats' spatial navigation skills or their ability to learn and remember a spatial navigation assignment. The animals were trained to get to a platform in a 180 cm by 60 cm circular pool with water kept at $28 \pm 2^\circ$ degrees Celsius and 40 cm deep. The four beginning locations on the pool's wall—A, B, C, and D—marked the division into four quadrants. A 9-cm-diameter circular platform was positioned on a column 2 cm below the water's surface inside the pool. Throughout the experiment, this platform—positioned in the middle of one of the four quadrants—acted as the only means of escaping from the water. A coloured flag was placed outside the pool at a consistent distance from

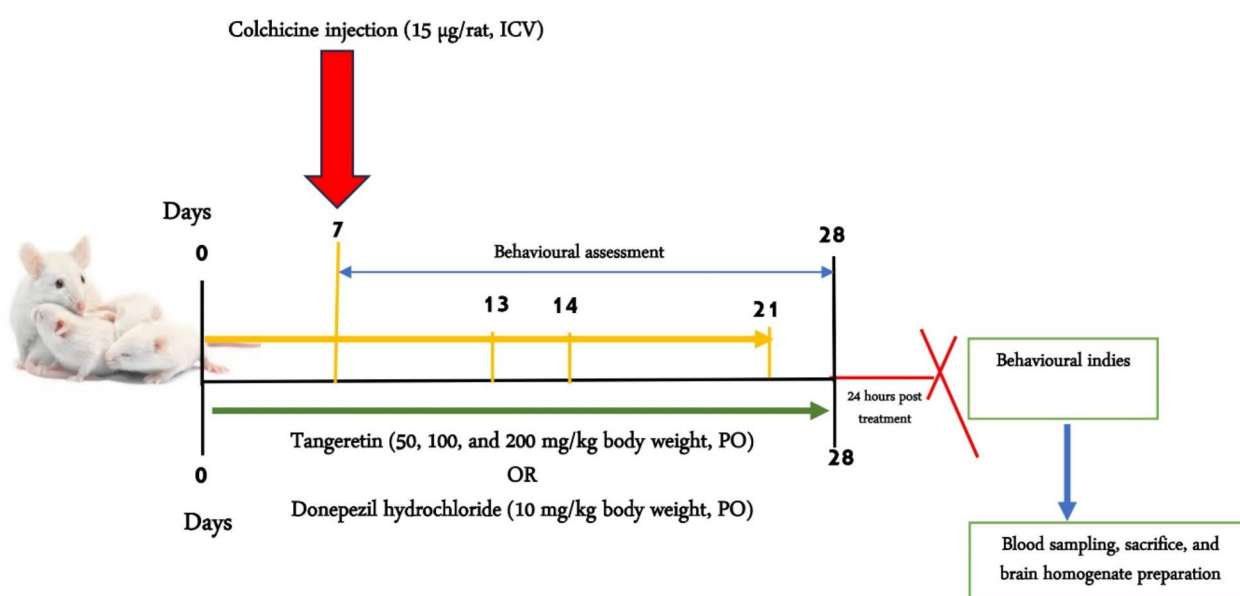


Fig. 1 Treatment protocol for rats exposed to physiological saline, colchicine, donepezil hydrochloride, or tangeretin

the pool to help the rat locate the platform submerged beneath the water [27].

(a) **Maze acquisition phase (training)**: Rats had four trials, separated by ten minutes, during a training session on the thirteenth day after the ICV injection. The four trials all started from different positions. The initial acquisition latency (IAL) was measured as the time a rat took to find the platform, with a maximum of 120 s (s).

(b) **Maze retention phase (testing for retention of the learned task)**: To assess retention of the previously acquired task, each rat was randomly placed at one of the starting sites (A–B–C–D), facing the pool wall, 24 h (day 14) and 8 days (day 21) after IAL. Days 14 and 21's journey times to the hidden platform were timed, and the results were labelled as the first and second retention latencies, or 1st and 2nd RL, respectively.

Assessment of gross behavioural activity

Rats' spontaneous motor activity and overall behaviour were assessed in an open field arena on days 13, 14, and 21 following the ICV injection. The wood-built, open-field chamber was 72 cm by 72 cm and 36 cm high. It had a glass wall and a floor divided into squares of the same size. The open-field challenge immediately preceded the water maze exam [28].

Biochemical tests

Biochemical evaluations were conducted on day 22, 24 h after the last behavioural testing, and on day 21, after the colchicine injections. The animals were euthanised by injection of an intraperitoneal dose of sodium pentobarbital (45 mg/kg), and after their brains were removed, they were washed in cold, isotonic saline. To homogenise brain tissue samples, ten times as much ice-cold 0.1 M phosphate buffer was used (with a pH of 7.4). After the final homogenate was centrifuged for 15 min at 10,000 × g, some of the supernatant was extracted and used for biochemical analysis.

Oxidative stress markers

Hydrogen peroxide generation The method described by Wolff was followed to determine the hydrogen peroxide level [29]. After the sample was added, the reaction mixture, known as FOX 1, was made up of 10 millilitres of 100 mM xylenol orange, 50 millilitres of 250 mM ammonium ferrous sulfate, 10 millilitres of 100 mM sorbitol, 5 millilitres of 25 mM H₂SO₄, and 30 millilitres of distilled water. Following a half-hour incubation period at room temperature, the absorbance at 560 nm was determined. The data were represented as micromoles per milligram

of protein and were calculated by extrapolating from the standard curve.

Total thiol content The method described by Ellman was followed to determine the total thiol content [30]. Exactly 20 µL of sample homogenate, 510 µL of phosphate buffer (pH 7.4, 0.1 M), 35 µL of DTNB (1 mM), and 35 µL of distilled water were combined to create a reaction mixture. After 30 min, the process was stopped at room temperature, and the absorbance at 412 nm was then measured.

Measurement of lipid peroxidation According to earlier protocols, the amount of malondialdehyde (MDA) in the rats' brains was measured. Exactly 300 µL of brain homogenate was mixed with 30% TCA and 5 N HCl, and then 2% TBA was added in 0.5 N NaOH. After 15 min of heating at 90 °C, this mixture was centrifuged for 10 min at 12,000 g. Using a UV-visible spectrophotometer, the pink colour of the resultant supernatant was detected at 532 nm. Using a standard curve made with tetra ethoxy propane, the concentration of MDA was calculated and expressed as nmol/mg of protein [31].

Levels of reactive oxygen species (ROS) Bondy and Guo's technique served as the foundation for determining ROS. Specifically, 2,7-dichlorofluorescein diacetate (DCFH-DA) is hydrolysed in homogenates to 2,7-dichlorohydrofluorescein (DCFH), which is further oxidised by reactive oxygen species (ROS) to fluorescent 2,7-dichlorofluorescein (DCF). In summary, 10 µL of homogenate and 10 µL of 1.25 M DCFH-DA dissolved in ethanol were added to 990 µL of 0.1 M phosphate buffer (pH 7.4). The mixture was then exposed to light and incubated for 30 min at 37 °C. At wavelengths $A_{em} = 525$ nm and $A_{ex} = 488$ nm, the product's fluorescence was measured. The ROS was calculated using a standard curve containing 10 µM DCF and expressed in µmoles of DCF/min/mg protein [32].

Antioxidant enzymes and reduced glutathione

Glutathione S-transferase activity Glutathione S-Transferase activity in brain tissue was measured using the procedures outlined in the technique created by Habig and Jakoby. In this case, 1-chloro-2,4-dinitrobenzene was the substrate. Potassium phosphate buffer (20 µL, 0.25 M, pH 7.0), EDTA (2.5 mM, 25 µL), pure water (10.5 µL), and GSH (500 µL, 0.1 M) were combined to create a solution. After adding CDNB (10 µL, 25 mM) and the sample (20 µL) to the mixture, the reaction mixture was left for five minutes. Using a spectrophotometer, absorbance mea-

measurements were obtained at 340 nm every ten seconds [33].

Glutathione peroxidase activity The method for measuring glutathione peroxidase activity in brain tissue was established by Rotruck et al. and applied in this investigation. A mixture of sodium phosphate buffer (500 μ L), 10 mM sodium azide (100 μ L), 4 mM GSH (200 μ L), 2.5 mM H_2O_2 (100 μ L), a 50 μ L sample adjusted to 2.0 mL with distilled water, 0.3 M disodium hydrogen phosphate (4 mL), and DTNB reagent were added to the reaction to stop it (1 mL). This mixture was incubated at 37 °C for three minutes. Using a spectrophotometer, absorbance measurements were made at 412 nm, and glutathione peroxidase activity was reported in nmol per milligram of protein [34].

Glutathione reductase activity This study's evaluation of GR activity adhered to the procedure by Carlberg and Mannervik. Exactly 1.65 ml of 0.1 M phosphate buffer with a pH of 7.6, 0.1 ml of 0.5 mM EDTA, 0.1 ml of 0.1 mM NADPH, 0.05 ml of 1 mM oxidised GSH, and 0.1 ml of 10% brain homogenate made up the reaction mixture for this test. With a spectrophotometer set to 340 nm, the changes in absorbance in the reaction mixture—which represent the loss of NADPH at 25 °C—were measured. Using a molar extinction coefficient of $6.22 \times 10^3 \text{ M}^{-1}\text{cm}^{-1}$, the data were quantified and represented as nmol NADPH oxidised/min/mg protein [35].

Catalase activity The catalase activity in brain tissue was measured in this investigation using the methodology developed by Aebi. The reaction mixture contained 20 μ L of the sample, 180 μ L of 300 mM H_2O_2 , and 1800 μ L of 50 mM potassium phosphate buffer (pH 7.0) (diluted at a ratio of 1:50). The catalase activity was measured by tracking the steady decrease in H_2O_2 absorbance over two minutes at 240 nm. The results were represented as μ mol of H_2O_2 depleted per minute per milligram of protein [36].

Superoxide dismutase activity Superoxide dismutase activity (SOD) was measured using the procedure described by Misra and Fridovich. The foundation of this procedure is its capacity to prevent adrenaline autooxidation at pH 10.2 and 30 °C. The sample (20 μ L) and 0.05 M carbonate-bicarbonate buffer (2.5 mL, pH 10.2) made up the reaction mixture. After that, epinephrine (0.3 mL, 0.3 mM) was added and well combined. A spectrophotometer measured the absorbance at 480 nm for 150 s, spaced 30 s apart. The findings were presented as SOD activity in nmol per milligram of protein [37].

Glutathione (GSH) level The glutathione levels in brain tissue were measured in this investigation using the method by Jollow et al. After adding an equivalent volume of 4% sulfosalicylic acid to each sample aliquot, centrifugation at 10,000 rpm for 15 min at 4 °C was used to deproteinise the samples. After that, DTNB was combined with 50 μ L of the acquired supernatants (10 mM, 4.5 mL). At a wavelength of 412 nm, the absorbance was measured, and the results were represented as μ mol/mg protein [38].

Neuroinflammatory markers

Cytokines level Cytokines such as TNF- α , NF- κ B, IL-1 β , IL-6, and IL-10 ELISA kits by the manufacturer's The levels of these inflammatory mediators and cytokines in brain tissue were measured in picograms per millilitre (pg/mL).

Nitrite level Using the Griess reagent (0.1% N-(1-naphthyl) ethylenediamine dihydrochloride, 1% sulfanilamide, and 2.5% phosphoric acid), the nitrite levels in the brain homogenate were determined, adhering to the protocol described by Green et al. The treated tissue sample was mixed with an equivalent amount of Griess reagent, and the absorbance was measured at 542 nm using a UV-visible spectrophotometer. A sodium nitrite standard curve was used to calculate the nitrite concentration, which was then expressed as μ g/mg of protein [39].

Myeloperoxidase (MPO) activity The method outlined by Granell et al. was used to assess the tissue's myeloperoxidase (MPO) activity. The reaction mixture contained 70 mL of diluted 3 mM H_2O_2 in 80 mM phosphate buffer, 10 mL of 1.6 mM tetramethylbenzidine dissolved in dimethyl sulfoxide, and 20 mL of homogenate supernatant (pH 5.4). Using a spectrophotometer, the absorbance was measured at 630 nm, and the activity of MPO was calculated as μ mol of H_2O_2 depleted per minute per milligram of protein [40].

Neurotransmitters and acetylcholinesterase (AChE) activity

Estimation of serotonin The method used to quantify serotonin in brain tissue was according to the protocol by Schlumpf et al. First, 0.25 mL of O-phthaldialdehyde (20 mg in 100 mL concentrated HCl) was added to 0.2 mL of the supernatant from the brain tissue. The mixture was heated to 100 °C for 10 min to create the fluorophore. With a spectrophotometer, measurements were made between 360 and 470 nm after the samples were allowed to reach equilibrium at room temperature. Exactly 0.25 mL of strong HCl without O-phthaldialdehyde was applied to the serotonin tissue blank. A 1:1 combination of distilled water and HCl-butanol generated an internal standard

serotonin (500 µg/ml). The serotonin (5-HT) concentration was calculated in ng per gram of tissue [41].

Estimation of glutamate and GABA The brain tissue supernatant was homogenised, and around 1.0 mL was dried in an oven at 70 °C. The remaining material was then reconstituted in 100 mL of distilled water. Using a micro-pipette, standard solutions of GABA at a concentration of 2 mM and glutamate (2.942 mg in 10 mL distilled water) were put as spots on the Whatman No. 1 chromatography paper beside the sample. After that, the paper was placed in a chamber with a solvent mixture of water, acetic acid, and butanol (12:3:5 v/v). The paper was taken out and dried once the solvent reached its top. A second run was conducted. Following drying, the papers were sprayed with a 0.25% ninhydrin reagent and baked for four minutes at 100 °C. The portions of the paper that, according to the standard, corresponded to glutamate were cut out and cleaned using 0.005% CuSO₄ in 75% ethanol. Using a spectrophotometer, their absorbance was calculated at 515 nm about a blank [41].

Estimation of dopamine Rat brain tissue was weighed and homogenised for about a minute in a 1:10 HCl-butanol mixture. The resultant sample was centrifuged at 3000 rpm for 10 min. One millilitre (mL) of the supernatant phase was removed and put in a centrifuge tube with 2.5 millilitres of hexane and 0.3 millilitres of 0.1 M HCl. This allowed for the aqueous phase (0.2 mL) extraction for dopamine measurement. Every operation was carried out at 0 °C (on ice). During the procedure, 0.1 mL of sodium acetate buffer (pH 6.9), 0.05 mL of 0.4 M HCl, and 0.2 mL of the aqueous phase were added. Then, 0.1 mL of iodine solution (0.1 M in ethanol) was added for oxidation. After two minutes, 0.1 mL of sodium sulphite solution was added to stop the reaction. After 1.5 min, 0.1 mL of acetic acid was added. For six minutes, the entire solution was heated to 100 °C. A spectrophotometer measured the excitation and emission spectra at 330–375 nm after the sample was warmed to room temperature. Tissue blanks were made for the dopamine estimate by adding the oxidation step reagents in the opposite order (sodium sulphite before iodine). The final dopamine assessment is expressed in pg/mg of tissue [41].

Acetylcholinesterase (AChE) activity Using the method by Ellman, the quantitative assessment of acetylcholinesterase activity in the brain was carried out. Acetylcholinesterase activity was measured to detect a decrease in cholinergic neurons. The supernatant (0.05 mL), 3 mL of 0.01 M sodium phosphate buffer (pH=8), 0.10 mL of acetylthiocholine iodide, and 0.10 mL of DTNB made up the test combination (Ellman reagent). At intervals of

15 s, the enzyme activity was measured at 412 nm with a UV-visible spectrophotometer. The specific activity of AChE was presented in nmoles/min/mg protein [42].

Apoptosis marker

Caspase-3 activity Caspase-3 activity in the tissues was assessed using a Caspase-3 ELISA kit from the manufacturer, strictly adhering to their specified procedural instructions. A microplate reader was used to measure absorbance at a wavelength of 450 nm, and the outcomes were presented as µmol/min/mg protein.

Neurotrophic factors

Estimation of brain-derived neurotrophic factor (BDNF) The Brain-Derived Neurotrophic Factor (BDNF) levels in the supernatant were measured following the manufacturer's instructions using an ELISA kit [43].

Levels of amyloid Beta (ab) peptide The Aβ peptide levels in the supernatant were measured following the manufacturer's instructions using an ELISA kit [44].

Protein estimation

Protein content was determined by Lowry et al. [45] method.

Analysis of data

Results are presented as Mean ± Standard Deviation (SD). The analysis of behavioural assessments involved a repeated measure two-way analysis of variance (ANOVA). Biochemical estimations were subjected to individual one-way ANOVA analyses. Subsequent *post hoc* comparisons between groups were conducted utilising Tukey's test. Significance was established at $p < 0.05$ with the aid of GraphPad Prism 5.0.

Results

Tangeretin improves colchicine-induced alteration in cognitive performance and gross behavioural activity in Wistar rats

The effects of colchicine on cognitive performance and behavioural activity in Wistar rats are given in Table 1. Unlike the colchicine-treated Wistar rats, rats treated with tangeretin and donepezil hydrochloride demonstrated rapid learning to navigate directly to the platform in the Morris water maze. Relative to the control group, colchicine at 15 µg/rat significantly ($p < 0.05$) decreased the cognitive performance and behavioural activity in the rats (by an increase in initial acquisition latency gradually with ongoing training during the spatial navigation task on day 13). Tangeretin and donepezil hydrochloride significantly ($p < 0.05$) increased the rats'

Table 1 Effect of tangeretin on the cognitive performance in Morris water maze (seconds) of Wistar rats with colchicine-induced memory impairment

	Mean transfer latency (s)		
	IAL (day 13)	1st RL (day 14)	2nd RL (day 21)
Physiological saline	58.77 ± 1.34	40.22 ± 0.91	30.57 ± 0.98
Colchicine + physiological saline	90.24 ± 1.35*	82.53 ± 1.22*	80.76 ± 1.02*
Colchicine + Donepezil hydrochloride (10 mg/kg)	66.78 ± 1.56 [#]	50.77 ± 0.98 [#]	45.23 ± 0.98 [#]
Colchicine + Tangeretin (50 mg/kg)	81.73 ± 1.67 [#]	61.75 ± 1.01 [#]	55.67 ± 1.11 [#]
Colchicine + Tangeretin (100 mg/kg)	68.77 ± 1.04 [#]	58.19 ± 1.13 [#]	50.87 ± 1.01 [#]
Colchicine + Tangeretin (200 mg/kg)	65.56 ± 1.03 [#]	48.93 ± 1.03 [#]	40.33 ± 1.01 [#]

s, seconds; IAL, initial acquisition latency; 1st RL, first retention latency; 2nd RL; second retention latency

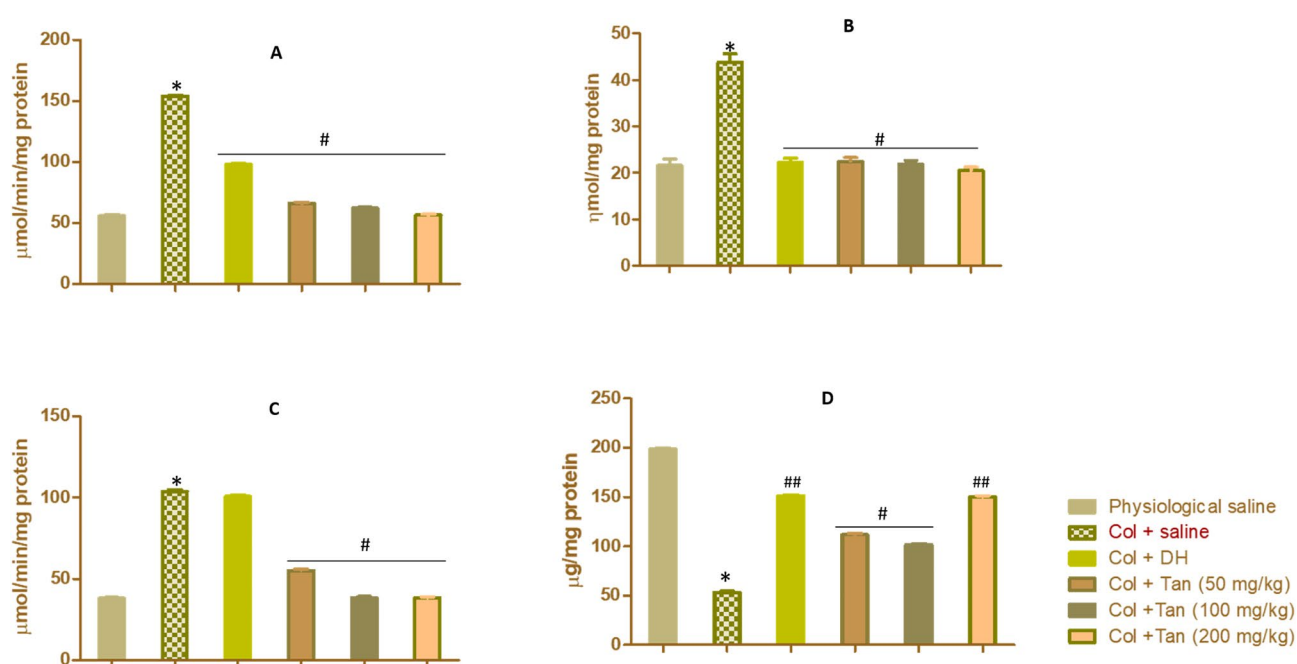


Fig. 2 Influence of tangeretin on some oxidative stress parameters in the brain of colchicine-induced memory impairment of male Wistar rats. H_2O_2 (A), MDA (B), ROS (C); depletion of T-SH (D) of male Wistar rats administered physiological saline, Colchicine + physiological saline, Colchicine + Donepezil hydrochloride, and Colchicine + Tangeretin (50 mg/kg, 100 mg/kg, and 200 mg/kg body weight). The values are presented as mean $\pm$ standard deviation ($n = 10$). Significant variations from the positive control group (physiological saline) are represented by * and from Colchicine + saline group by #, ($p < 0.05$)

cognitive performance and behavioural activity. Doses of tangeretin significantly ($p < 0.05$) decreased the colchicine-induced increase of initial acquisition latency (day 13), 1st RL (day 14), and 2nd RL (day 21) relative to the colchicine-treated group, which compared well with the group treated with physiological saline.

Tangeretin ameliorates colchicine-induced accrue ment of H_2O_2 , MDA, ROS; depletion of T-SH in Wistar rats

The results from the exposure of Wistar rats to colchicine and tangeretin on selected oxidative stress biomarkers are shown in Fig. 2. Colchicine at 15 $\mu\text{g}/\text{rat}$ ($p < 0.05$) significantly altered the H_2O_2 , MDA, and T-S depletion compared to the control group. When compared with the control, the alterations caused by colchicine were reversed in H_2O_2 level (up to 2.3 folds, 2.5 folds and 2.7 folds reduction by 50, 100, and 200 mg/kg respectively by

tangeretin and 1.57 folds reduction by donepezil hydrochloride, Fig. 2A); level of MDA (up to 1.9 folds, 2.2 folds and 2.1 folds reduction by 50, 100, and 200 mg/kg respectively by tangeretin and 2.0 folds reduction by donepezil hydrochloride, Fig. 2B); level of ROS (up to 1.9 folds, 2.7 folds and 2.7 folds reduction by 50, 100, and 200 mg/kg respectively by tangeretin and 1.0 folds reduction by donepezil hydrochloride, Fig. 2C); and T-SH level (up to 2.1 folds, 1.9 folds and 2.8 folds elevation by 50, 100, and 200 mg/kg respectively by tangeretin and 2.8 folds elevation by donepezil hydrochloride, Fig. 2D).

Tangeretin reversed colchicine-induced inhibition of catalase, SOD, GPX, GR and GST activities; and GSH in Wistar rats

The effect of treatment of the Wistar rats with colchicine and tangeretin on selected antioxidant biomarkers is

presented in Fig. 3. The results indicated that colchicine at 15 μ /rat ($p < 0.05$) significantly inhibited the activities of the antioxidant enzymes and reduced glutathione level when compared with the physiological saline-treated Wistar rats (Fig. 3). When compared with the control, the reductions caused by colchicine were reversed in catalase activity (46%, 47% and 50% increase by 50, 100, and 200 mg/kg respectively by tangeretin and 45% increase by donepezil hydrochloride, Fig. 3A); SOD activity (45%, 50% and 54% increase by 50, 100, and 200 mg/kg respectively by tangeretin and 54% increase by donepezil hydrochloride, Fig. 3B); GPx activity (51%, 53% and 52% increase by 50, 100, and 200 mg/kg respectively by tangeretin and 45% increase by donepezil hydrochloride, Fig. 3C); GR activity (36%, 38% and 41% elevation by 50, 100, and 200 mg/kg respectively by tangeretin and 39% elevation by donepezil hydrochloride, Fig. 3D); GST activity (50%, 50% and 50% increase by 50, 100, and 200 mg/kg respectively by tangeretin and 50% increase by donepezil hydrochloride, Fig. 3E); GSH level by tangeretin (66%, 77% and 66% elevation by 50, 100, and 200 mg/kg respectively, or donepezil hydrochloride with 65% elevation, Fig. 3F).

Tangeretin reversed colchicine-induced increment of TNF- α , NF- κ B, IL-1 β , IL-6, IL-10, and nitrite levels as well as myeloperoxidase activity in Wistar rats

The effect of the exposure of Wistar rats to colchicine and tangeretin on selected cytokines and neuroinflammatory markers is indicated in Fig. 4. The 15- μ g dose of colchicine ($p < 0.05$) significantly increased the parameters when compared with the control animals treated with physiological saline (Fig. 4). However, compared to colchicine treated group, tangeretin at 50, 100 and 200 mg/kg, as well as the 10 mg/kg of donepezil hydrochloride, significantly ($p < 0.05$) reversed colchicine-induced elevation of TNF- α level (68%, 70%, 71% and 70% reduction respectively, Fig. 4A), NF- κ B level (75%, 76%, 78% and 75% reduction respectively, Fig. 4B), IL-1 β level (62%, 63%, 64% and 61% reduction respectively, Fig. 4C), IL-6 level (61%, 62%, 62% and 70% reduction respectively, Fig. 4D), IL-10 level (73%, 74%, 75% and 72% reduction respectively, Fig. 4E), nitrite level (64%, 61%, 66% and 46% reduction respectively, Fig. 4F), and myeloperoxidase activity (72%, 78%, 80% and 65% reduction respectively, Fig. 4G) in Wistar rats. The 100 mg/kg of tangeretin compared favourably well with the control groups treated with physiological saline.

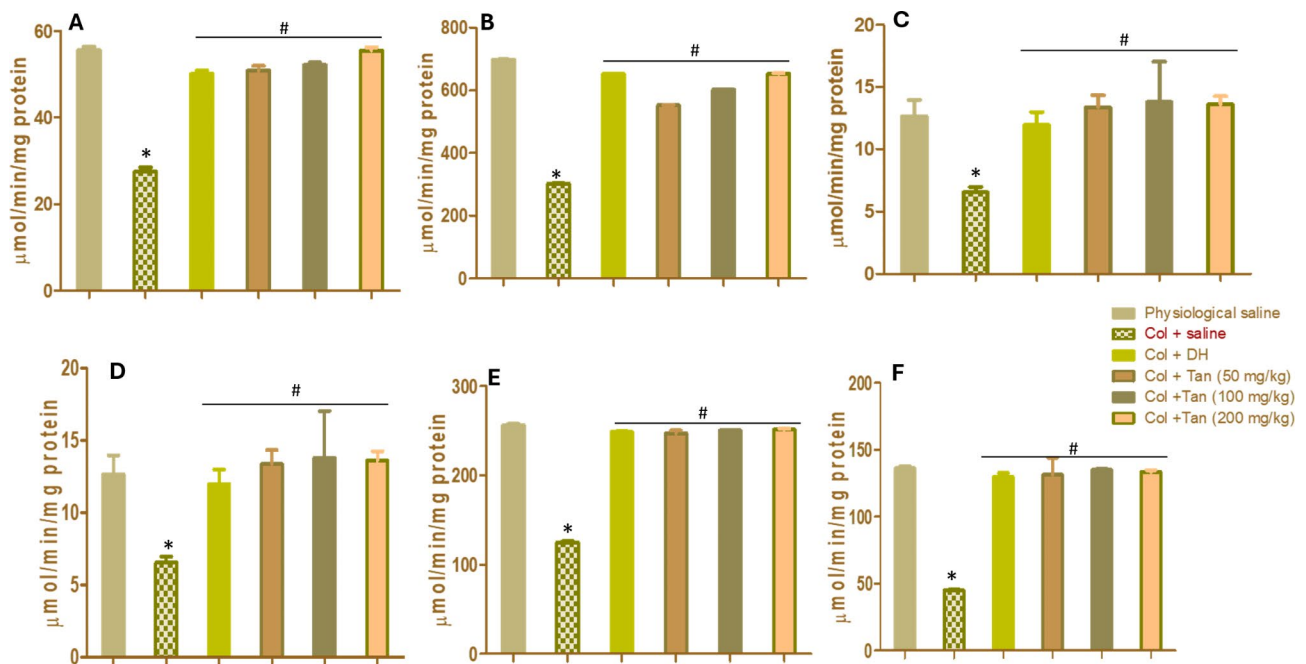


Fig. 3 Efficacy of tangeretin on antioxidant status in the brain of Wistar rats with colchicine-induced memory impairment. Catalase (A), SOD (B), GPx (C), GR (D) and GST (E) activities; and GSH (F) of male Wistar rats administered physiological saline, Colchicine + physiological saline, Colchicine + Donepezil hydrochloride, and Colchicine + Tangeretin (50 mg/kg, 100 mg/kg, and 200 mg/kg body weight). The values are presented as mean $\pm$ standard deviation ($n = 10$). Significant variations from the positive control group (physiological saline) are represented by * and from Colchicine + saline group by #, ($p < 0.05$)

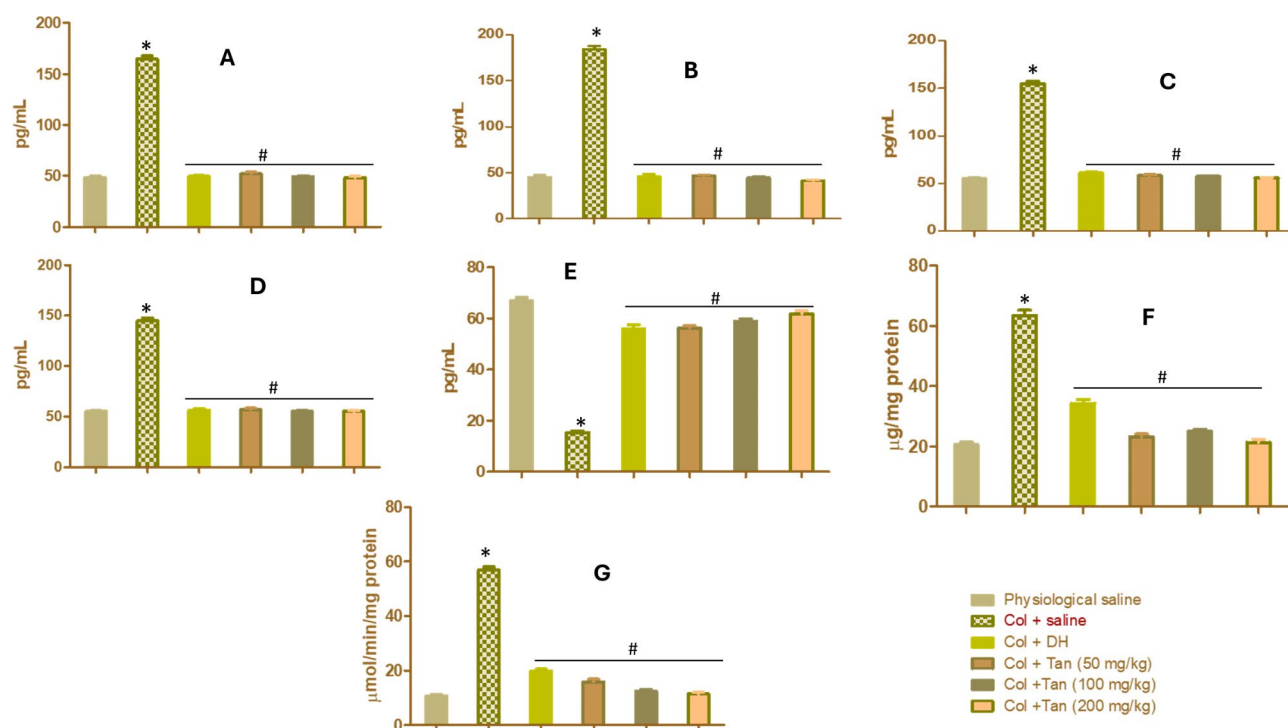


Fig. 4 Impact of tangeretin on neuroinflammatory markers in the brain of Wistar rats induced to memory impairment with colchicine. TNF- α (A), NF- κ B (B), IL-1 β (C), IL-6 (D), IL-10 (E), and nitrite levels (F), as well as myeloperoxidase activity (G) of male Wistar rats, administered physiological saline, Colchicine + physiological saline, Colchicine + Donepezil hydrochloride, and Colchicine + Tangeretin (50 mg/kg, 100 mg/kg, and 200 mg/kg body weight). The values are presented as mean $\pm$ standard deviation ($n = 10$). Significant variations from the positive control group (physiological saline) are represented by * and from Colchicine + saline group by #, ($p < 0.05$)

Tangeretin refurbish colchicine-induced alterations of serotonin (5-HT), dopamine, GABA and glutamate levels as well as activity of AChE in Wistar rats

The effect of the exposure of Wistar rats to colchicine and tangeretin on selected neurotransmitters and the activity of AChE is indicated in Fig. 5. Our findings revealed that colchicine at 15 μg resulted in significant ($p < 0.05$) alterations in the levels of neurotransmitters and activity of AChE when compared to the control group (Fig. 5). When compared with the control, the alterations caused by colchicine were reversed in serotonin levels (45%, 54% and 58% reduction by 50, 100, and 200 mg/kg respectively by tangeretin and 45% reduction by donepezil hydrochloride, Fig. 5A); level of dopamine (51%, 52% and 52% reduction by 50, 100, and 200 mg/kg respectively by tangeretin and 51% reduction by donepezil hydrochloride, Fig. 5B); level of glutamate (61%, 64% and 66% reduction by 50, 100, and 200 mg/kg respectively by tangeretin and 62% reduction by donepezil hydrochloride, Fig. 5C); level of GABA (52%, 53% and 54% elevation by 50, 100, and 200 mg/kg respectively by tangeretin and 51% elevation by donepezil hydrochloride, Fig. 5D); and AChE activity by tangeretin (63%, 57% and 58% reduction by 50, 100, and 200 mg/kg respectively, Fig. 5E) or donepezil hydrochloride (56% reduction).

Tangeretin ameliorate colchicine-induced alterations of caspase-3 activity and levels of amyloid beta (ab) peptide and BDNF in Wistar rats

Results herein revealed that colchicine at 15 $\mu\text{g}/\text{rat}$ resulted in a significant ($p < 0.05$) decrease in the level of BDNF but increased the level of Ab peptide and activity of caspase-3 when relative to the control group (Fig. 6). When compared with the control, the alterations caused by colchicine were reversed in BDNF level (up to 2.0 folds, 2.0 folds and 2.0 folds elevation by 50, 100, and 200 mg/kg respectively by tangeretin and 1.8 folds elevation by donepezil hydrochloride, Fig. 6A); level of amyloid beta (Ab) peptide (up to 3.8 folds, 4.4 folds and 4.3 folds reduction by 50, 100, and 200 mg/kg respectively by tangeretin and 2.8 folds reduction by donepezil hydrochloride, Fig. 6B); and caspase-3 activity by tangeretin (up to 2.6 folds, 3.5 folds and 5.5 folds reduction by 50, 100, and 200 mg/kg respectively, Fig. 6C) or donepezil hydrochloride (4.0 folds reduction).

Discussion

Presently, there is no cure for Alzheimer's disease, and available treatments only provide symptomatic relief without altering the course of the disease. Several research aim to uncover the underlying causes and mechanisms of AD, paving the way for the development

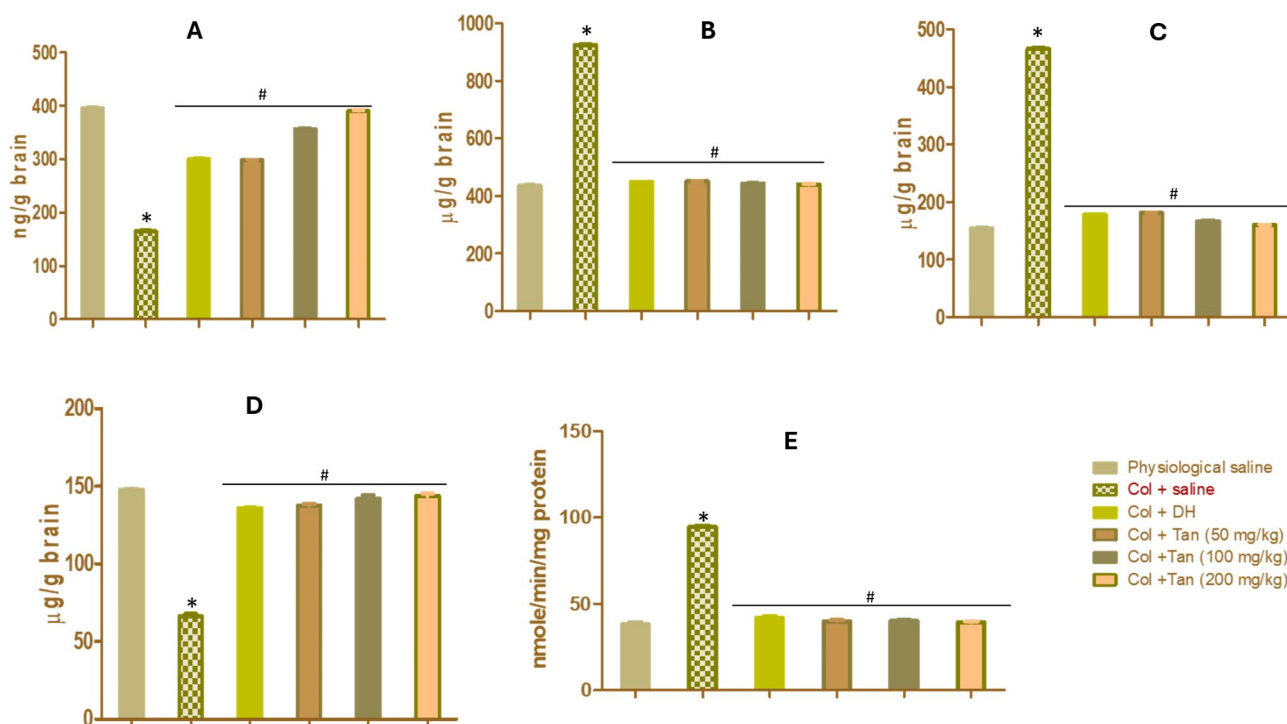


Fig. 5 Capacity of tangeretin on neurotransmitters and acetylcholinesterase activity in the brain of colchicine-induced memory impairment of male Wistar rats. Serotonin (**A**), dopamine (**B**), GABA (**C**) and glutamate (**D**) as well as activity of AChE (**E**) of male Wistar rats administered physiological saline, Colchicine + physiological saline, Colchicine + Donepezil hydrochloride, and Colchicine + Tangeretin (50 mg/kg, 100 mg/kg, and 200 mg/kg body weight). The values are presented as mean $\pm$ standard deviation ($n = 10$). Significant variations from the positive control group (physiological saline) are represented by * and from Colchicine + saline group by #, ($p < 0.05$)

of disease-modifying treatments that could slow or halt its progression. Research on Alzheimer's disease contributes to a broader understanding of neurodegenerative processes in the brain. Insights gained from studying AD may have implications for other neurodegenerative disorders, potentially leading to breakthroughs in the treatment of conditions such as Parkinson's disease and Huntington's disease.

In the current study, colchicine was found to impair learning and memory while also causing damage to central neurons. The decreased cognitive performance and behavioural activity in rats, as indicated by an increase in initial acquisition latency (IAL) during the spatial navigation task with ongoing training by colchicine, suggest significant impairments in learning and memory processes. Moreover, in the Morris water maze task, rats induced with colchicine showed a substantial increase in IAL and in the first and second retention latencies (1st and 2nd RL) to reach the platform compared to control rats. These findings indicate a disruption in cognitive function, likely stemming from interference with critical neuronal microtubule dynamics essential for synaptic plasticity and memory formation [27, 46]. Conversely, tangeretin significantly improved retention performance in the spatial navigation task of the Morris water maze. This improvement was evidenced by the significant

decrease in IAL, 1st, and 2nd RL, suggesting its potential as a therapeutic agent for cognitive enhancement and neuroprotection.

The biochemical parameters in this study provide insights into the mechanisms through which tangeretin may exert its neuroprotective effects and mitigate colchicine-induced memory impairment in Wistar rats. The impact of tangeretin on oxidative stress, neuroinflammation, apoptosis, neurotransmitter balance, and overall neuronal health can be assessed by measuring these parameters. For instance, oxidative stress markers like H_2O_2 generation, total thiol content, level of malondialdehyde; antioxidant enzymes like glutathione S-transferase, glutathione peroxidase, glutathione reductase, catalase, superoxide dismutase; cytokines such as TNF- α , NF- κ B, IL-1 β , IL-6, and IL-10; neuroinflammatory parameters like nitrite, myeloperoxidase; neurotransmitters (serotonin, dopamine, glutamate, and GABA) and acetylcholinesterase; neurotrophic factors (Brain-Derived Neurotrophic Factor, A β - peptide) are significant contributors to a broader understanding of neurodegenerative processes in the brain.

Oxidative stress has been implicated in various neurodegenerative disorders, including those associated with memory impairment, such as Alzheimer's disease [47]. Excessive generation of hydrogen peroxide can lead to

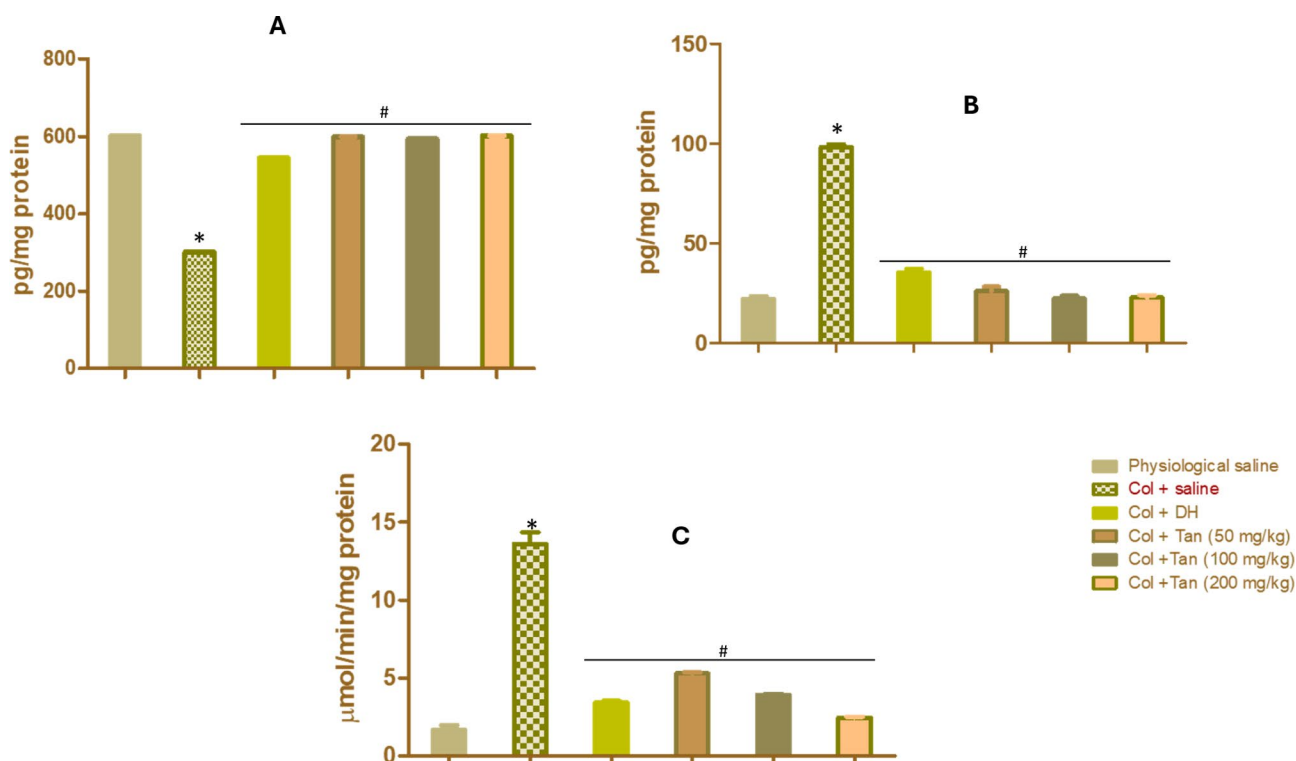


Fig. 6 Effects of tangeretin on apoptotic marker and neurotrophic factors in the brain of male Wistar rats with colchicine-induced memory impairment. Caspase-3 activity (A), and levels of amyloid beta peptide (B) and BDNF (C) of male Wistar rats administered physiological saline, Colchicine + physiological saline, Colchicine + Donepezil hydrochloride, and Colchicine + Tangeretin (50 mg/kg, 100 mg/kg, and 200 mg/kg body weight). The values are presented as mean $\pm$ standard deviation ($n = 10$). Significant variations from the positive control group (physiological saline) are represented by * and from Colchicine + saline group by #, ($p < 0.05$)

oxidative damage to lipids, proteins, and nucleic acids in the brain, ultimately impacting synaptic plasticity, neuronal communication, and memory formation [47, 48]. In contrast, oxidative stress can lead to the depletion of thiol antioxidants, compromising the cell's ability to counteract oxidative damage and contributing to neuronal dysfunction and cognitive decline [47, 49]. Lipid peroxidation-mediated damage to neuronal membranes can disrupt cellular integrity, impair neurotransmission, and contribute to neuronal death, ultimately leading to cognitive dysfunction. In contrast, ROS-mediated oxidative damage to biomolecules, including proteins, lipids, and nucleic acids, can disrupt cellular function and neuronal signalling pathways critical for memory formation and maintenance [47, 50]. Colchicine is a microtubule-disrupting agent that causes rodent cognitive impairment via mechanisms involving oxidative stress and altered antioxidant defence systems [51]. The increase in hydrogen peroxide generation, depletion of total thiol content and increase in lipid generation elicited by colchicine in this study suggests oxidative stress, a condition characterised by an imbalance between the production of reactive oxygen species (ROS) and the ability to neutralise them through antioxidant defences. These results further suggest oxidative damage in neuronal cells, affecting

cellular structures and impairing the proper functioning of the brain [52]. The reversal of these oxidative stress markers by tangeretin suggests its ability to neutralise reactive oxygen species (ROS) and free radicals, reducing oxidative damage to cells, including neurons in the brain [50]. This may be associated with its ability to stimulate the activity of endogenous antioxidant enzymes, such as superoxide dismutase (SOD), catalase, and glutathione peroxidase enzymes that play a crucial role in the cellular defence against oxidative stress, which was indeed enhanced in this study [53]. Tangeretin might have targeted lipid peroxidation and MDA accumulation, such as antioxidants or inhibitors of lipid peroxidation pathways, which may hold promise for ameliorating memory impairment associated with oxidative stress [20]. Also, Tangeretin, unlike donepezil hydrochloride in this study, might have actively modulated oxidative stress pathways and restored redox balance, suggesting its therapeutic use for preserving cognitive function and mitigating memory deficits in these conditions.

GSTs are a family of enzymes that catalyse the conjugation of glutathione with electrophilic compounds, facilitating their detoxification and excretion, and GPx enzymes are responsible for catalysing the reduction of hydrogen peroxide and organic hydroperoxides using

glutathione as a cofactor, thus protecting cells from oxidative damage. GR is an enzyme that catalyses the reduction of oxidised glutathione (GSSG) back to its reduced form (GSH), which is essential for maintaining cellular antioxidant defences. GSH is a tripeptide (γ -glutamyl-cysteinyl-glycine) critical cellular antioxidant and cofactor for various antioxidant enzymes [54]. CAT is an antioxidant enzyme that catalyses the decomposition of hydrogen peroxide into water and oxygen, thereby preventing the accumulation of this reactive oxygen species [55]. SOD enzymes catalyse the dismutation of superoxide radicals ($O_2^{\cdot-}$) into hydrogen peroxide and molecular oxygen, thereby preventing the formation of highly reactive hydroxyl radicals [56]. Dysregulation of these antioxidant enzymes and molecules, including GST, GPx, GR, CAT, SOD, and GSH, is commonly observed in memory impairment and neurodegenerative diseases. These alterations contribute to oxidative stress, disrupted redox signalling, and neuronal damage, ultimately leading to cognitive dysfunction [57]. Therapeutic interventions aimed at restoring antioxidant balance and reducing oxidative stress may hold promise for preserving cognitive function and ameliorating memory deficits in these conditions. It's worth noting that while oxidative stress is a recognised factor in memory-related conditions, it is often part of a complex multifactorial process involving other mechanisms like inflammation and mitochondrial dysfunction [50]. A decrease in antioxidant enzymes is a common observation in conditions associated with memory impairment, particularly in neurodegenerative disorders like Alzheimer's disease [58]. In this study, the decrease in the activity of glutathione S-transferase, glutathione peroxidase, glutathione reductase, catalase, and superoxide dismutase by colchicine suggests that the role of these antioxidant enzymes in protecting cells, including neurons in the brain, from oxidative stress by neutralising reactive oxygen species (ROS) and free radicals have been compromised [59]. The decline in antioxidant enzyme levels is often linked to the overall increase in oxidative stress observed in conditions like Alzheimer's disease [47]. Therefore, the increase in oxidative stress markers and the decrease in the antioxidant enzymes in this study lay more credence to oxidative damage, a contributing factor in the progression of memory-related disorders [59, 60]. This suggests that colchicine disrupted the balance of antioxidant enzymes, leading to oxidative stress and memory impairment. The ability of tangeretin to have reversed the level of oxidative stress markers while restoring the activities of the antioxidant enzymes and glutathione levels in this study indicates its efficacy in halting the progression of memory-related disorders [16, 61]. The findings here suggests that tangeretin can potentially counteract colchicine-induced memory impairment by restoring the balance of antioxidant enzymes. These

findings lay more credence to the therapeutic potentials of tangeretin at restoring antioxidant balance and reducing oxidative stress, thereby holding promise for preserving cognitive function and ameliorating memory deficits in these conditions.

Furthermore, memory impairment conditions, particularly neurodegenerative disorders like Alzheimer's disease, are associated with alterations in neurotransmitter systems. This implies that neurotransmitter dysregulation is critical in memory and cognition [62, 63]. Dopamine is a neurotransmitter that plays a crucial role in various cognitive processes, including memory formation and retrieval. In contrast, dopamine is known to enhance memory retrieval performance; it can also have a dual effect by impairing metacognition, which refers to the ability to monitor and control one's cognitive processes [64, 65]. Lower brain serotonin levels have been linked to dementia and memory decline [66]. Also, GABA and glutamate are critical players in maintaining the balance between excitatory and inhibitory neurotransmission, which is vital for normal brain function and memory processing [67]. Furthermore, AChE is an enzyme responsible for the hydrolysis of the neurotransmitter acetylcholine (ACh) into choline and acetate, terminating cholinergic neurotransmission [68]. In this study, colchicine's alterations in dopamine, serotonin, glutamate, and GABA suggest a bridge of communication between nerve cells (neurons) in the brain. At the same time, the marked increase in AChE activity reflects compensatory upregulation in response to cholinergic deficits since elevated AChE activity accelerates the breakdown of ACh, further exacerbating cholinergic neurotransmission impairment and contributing to memory deficits [67, 69]. Tangeretin could have possibly repaired the destruction of GABAergic and glutamatergic neurons (elicited by colchicine in causing memory impairment), thereby leading to the availability of these neurotransmitters, perhaps leading to memory impairment [70]. The increased dopamine and reduced serotonin levels by colchicine is a pointer to impaired metacognition, affecting confidence in decision-making during memory tasks. The tangeretin effect in this study in modulating the alterations might be associated with the modulating synaptic function to improve cognitive symptoms in memory impairment conditions by targeting neurotransmitter systems. An increase in serotonin function in the brain by tangeretin could have prevented memory loss from worsening [71, 72]. Furthermore, the ability of tangeretin, like that of donepezil hydrochloride, to enhance cholinergic transmission suggests its efficacy as a cholinesterase inhibitor in slowing disease progression, highlighting the need for novel therapeutic approaches targeting underlying neurodegenerative mechanisms [73].

The increase in the levels of TNF- α , NF- κ B, IL-1 β , IL-6, and IL-10 by colchicine suggests activation of microglia, which contributes to the release of inflammatory mediators denoting chronic inflammation within the central nervous system [74]. This implies that colchicine resulted in impaired memory, neurodegeneration, and changes in proinflammatory cytokine levels in the hippocampus and other brain regions in the experimental rats [75]. An increase in nitrite levels and the activity of myeloperoxidase (MPO) in memory-impaired rats can be indicative of neuroinflammation and oxidative stress, both of which are commonly associated with cognitive dysfunction and neurodegenerative diseases [76, 77]. In this study, colchicine led to excessive MPO activity and neutrophil infiltration, which can exacerbate neuroinflammation and neuronal injury, further impairing memory and cognitive function. Tangeretin and donepezil hydrochloride action on the proinflammatory cytokines in this study can be linked to its ability to modulate the inflammatory response in mitigating cognitive decline and improving the quality of life of Wistar rats with memory impairment conditions [78]. It may have acted by suppressing the activation of microglia, the immune cells in the brain that contribute to the release of proinflammatory cytokines [79]. Since oxidative stress is often linked to inflammation, and by reducing it, tangeretin may indirectly lower the proinflammatory cytokine levels in this study [80]. By protecting neurons, tangeretin may have also mitigated the signals that trigger the release of these proinflammatory cytokines. Tangeretin may also have anti-inflammatory effects, reducing the production of inflammatory mediators that contribute to oxidative damage in this study [80, 81]. Overall, the decrease in nitrite levels and MPO activity by tangeretin suggests the inhibition of neuroinflammation and oxidative stress, thereby preventing cognitive dysfunction and memory deficits [82]. Targeting these pathways, such as by inhibiting neuroinflammation or reducing oxidative stress, may represent potential therapeutic strategies for ameliorating memory impairment in neurodegenerative diseases and other cognitive disorders [83].

Brain-derived neurotrophic factor (BDNF) is a protein that plays a crucial role in neuronal survival, growth, and synaptic plasticity, particularly in regions of the brain involved in learning and memory, such as the hippocampus [84]. At the same time, the amyloid beta (A β) is a peptide derived from the proteolytic cleavage of the amyloid precursor protein (APP) [85]. In memory impairment and neurodegenerative diseases such as Alzheimer's disease (AD), the levels of BDNF and A β peptide are often dysregulated, reflecting underlying pathological processes that contribute to cognitive decline [86, 87]. In this study, an increase in A β peptide level by colchicine suggests its accumulation, particularly

in the form of insoluble aggregates known as amyloid plaques, is a hallmark pathological feature of AD, while the decrease in BDNF levels suggests impaired synaptic function, reduced neurogenesis, and compromised neuronal survival, all of which can contribute to memory deficits [86, 88]. On the other hand, tangeretin may have attenuated A β peptide accumulation and aggregation by inhibiting the formation of A β fibrils and reducing A β -induced neurotoxicity, possibly due to its antioxidant and anti-inflammatory properties, which may be crucial in mitigating the neurotoxic effects of A β peptides [89]. Furthermore, increased BDNF levels following tangeretin treatment may be associated with improved learning and memory performance in animal models of neurodegenerative diseases. By enhancing BDNF expression, tangeretin, like donepezil hydrochloride, may support neuronal survival, synaptic plasticity, and cognitive function in the brain [84, 90].

Caspase-3 is a cysteine protease that plays a central role in the execution phase of apoptosis, a programmed cell death process [91]. In memory impairment, including neurodegenerative diseases such as Alzheimer's disease (AD), alterations in the activity of specific enzymes such as Caspase-3 play crucial roles in the pathophysiology of the condition [92]. The increase in the activity of this enzyme by colchicine suggests cleavage of various cellular substrates, including structural proteins and DNA repair enzymes, resulting in apoptotic cell death, contributing to neuronal loss and synaptic dysfunction, ultimately impairing cognitive function and memory formation in this study [93]. On the other hand, tangeretin action similar to that of donepezil hydrochloride, to have lowered the activity of Caspase-3, can be linked to its efficacy in inhibiting caspase-3 activation or blocking downstream apoptotic pathways, which may hold promise for preserving neuronal integrity and ameliorating memory deficits in neurodegenerative diseases [94].

Conclusion

In conclusion, this study investigated the potential of tangeretin in alleviating cognitive dysfunction and oxidative stress induced by colchicine in rats, comparing its efficacy with donepezil hydrochloride. The findings revealed that colchicine administration led to significant cognitive impairment and increased oxidative stress markers in rat brains. Chronic treatment with tangeretin at doses of 50, 100, and 200 mg/kg orally once daily effectively ameliorated colchicine-induced cognitive deficits and reversed the altered biochemical parameters associated with oxidative stress. Notably, tangeretin demonstrated a protective role by reducing Caspase-3 activity and modulating various oxidative stress markers, neurotransmitters, inflammatory cytokines, and antioxidant enzymes. In contrast, donepezil hydrochloride did

not exhibit similar effects in preventing oxidative stress. These results highlight the potential therapeutic benefits of tangeretin in preserving cognitive function and combating oxidative stress in neurodegenerative conditions like Alzheimer's disease. Further research is warranted to explore the underlying mechanisms of tangeretin's neuroprotective effects and its implications for developing novel treatments for cognitive impairments associated with neurodegenerative diseases.

Abbreviations

AD	Alzheimer's disease
ICV	Intracerebroventricularly
H ₂ O ₂	Hydrogen peroxide
MPO	Myeloperoxidase
ROS	Nitrite reactive oxygen species
TNF- α	Tumour necrosis factor α
NF κ B	Nuclear factor kappa B
IL-1 β	Interleukin-1 β
IL-6	Interleukin-6
IL-10	Interleukin-10
BDNF	Brain derived neurotrophic factor
GABA	Gamma-aminobutyric acid
AChE	Acetylcholinesterase
GST	Glutathione S Transferase
GPx	Glutathione peroxidase
GR	Glutathione reductase
CAT	Catalase
SOD	Superoxide dismutase
GSH	Reduced glutathione
TSH	Total thiol
6-OHDA	6-Hydroxydopamine
ELISA	Enzyme Linked Immunosorbent Assay
BDNF	Brain Derived Neurotrophic Factor
Ab	Amyloid beta
NAS	National Academy of Science's
DCFH-DA	2,7-dichlorofluorescein diacetate
DCFH	2,7-dichlorohydrofluorescein
DCF	fluorescent 2,7-dichlorofluorescein ()

Acknowledgements

The authors sincerely thank the Pharmacology, Drug Discovery, and Toxicology Research Laboratory, Department of Biological Sciences, KolaDaisi University, Ibadan, Nigeria.

Author contributions

Conception and design, OBO; acquisition of data, OBO, MEK, GAG; analysis, interpretation of data and drafting of the manuscript, OBO, MEK, AT and AHA; revising for intellectual content, OBO, MEK and GAG. Final approval, OBO, MEK, GAG, AT and AHA. All authors read and agreed to publish the manuscript.

Funding

The research was not funded.

Data availability

Data and materials would be made available upon reasonable request.

Declarations

Ethics approval and consent to participate

The research procedure was approved by Ajayi Crowther University's research ethics committee (FNS/ERC/2022/020B) and followed the National Academy of Science's (NAS) recommendations for the moral use and appropriate handling of laboratory animals.

Consent for publication

All authors attested to the publication of the manuscript.

Competing interests

The authors declare no competing interests.

Clinical trial number

Not applicable.

Author details

¹Pharmacology, Drug Discovery and Toxicology Research Laboratory, Department of Biological Sciences, KolaDaisi University, Ibadan 200213, Nigeria

²Department of Biotechnology and Food Science, Faculty of Applied Sciences, Durban University of Technology, P.O. Box 1334, Durban 4000, South Africa

³Department of Pharmacology and Toxicology, College of Medicine, Taif University, Taif 21944, Kingdom of Saudi Arabia

⁴Oral Biology Department, Faculty of Dentistry, King Abdulaziz University, Jeddah, Saudi Arabia

Received: 10 August 2024 / Accepted: 17 January 2025

Published online: 04 February 2025

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